C-E-230E
September 28, 1977
SUPERSEDING
Int. Fed. Spec. C-E-00230D(Army-GL)
November 23, 1973 and
Fed. Spec. C-E-230C
November 16, 1970

FEDERAL SPECIFICATION

EGG AND EGG PRODUCTS, FROZEN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers frozen whole eggs, egg white, egg yolks, sugar yolks, and salt yolks for use by agencies of the Federal Government (see 6.2).

1.2 Classification. The product shall be of the following types and kinds as specified (see 6.1):

Types:

I - The storage time prior to delivery shall not exceed 180 days from date of production.
II - Shall be produced in a period not to exceed 90 days prior to date of delivery.

Kinds:

a - Frozen whole egg, table grade (type II).
b - Frozen whole eggs.
c - Frozen egg white.
d - Frozen egg yolk.
e - Frozen sugared yolk.
f - Frozen salted yolk.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 8910

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Federal Specifications:

QQ-T-425	-	Tinplate (Hot Dip and Electrolytic).
TT-C-495	-	Coatings, Exterior, for Tinned Food Cans.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-1163	-	Box, Corrugated Fiberboard, High Compression Strength, Weather Resistant, Wax-Resin Impregnated.
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.

Federal Standards:

FED-STD-101	-	Preservation, Packaging, and Packing Materials: Test Procedures.
FED-STD-123	-	Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Philadelphia, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtaind from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

US Department of Agriculture

Regulations Governing the Grading of Shell Eggs and United States Standards, Grades, and Weight Classes for Shell Eggs (7 CFR Part 56)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

United States Standards for Condition of Food Containers

USDA Laboratory Methods for Egg Products

(Application for copies should be addressed to the Poultry Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Method of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

American Public Health Association

Compendium of Methods for the Microbiological Examination of Foods

(Application for copies should be addressed to the American Public Health Association, 1015 18th Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Materials. The materials shall conform to the requirements of table I.

TABLE I. Material requirements

Item	Requirement
1	a. All products covered in this specification shall be produced from chicken shell eggs which have moved through usual marketing channels since the time they were laid and have not been held in refrigerated storage in excess of 20 days. b. For kind (a), the eggs shall meet the interior quality requirements for US Grade A or better, as described in The USDA Regulations Governing the Grading of Shell Eggs and United States Standards, Grades, and Weight Classes for Shell Eggs (7 CFR Part 56). Such interior quality shall have been determined within 10 days prior to the date of breaking (see 3.1.2). c. Alternatively, for kind (a) product, commercial nest-run eggs, meeting the age requirements of a., above, may be used, providing the conditions of 3.1.1 are met.

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TABLE I. Material requirements (cont'd)

Item	Requirement
2	For kind (a), prior to pasteurization, the raw material shall have an aerobic plate count not to exceed 250,000 per gram. 1/
3	Sugar shall be white, refined, granulated cane or beet sugar.
4	Salt shall be white, refined, granulated or flake-type sodium chloride. Iodized salt shall not be used.

1/ If eggs are shipped from a breaking plant to another plant for pasteurization and packaging, the aerobic plate count shall be determined at the second processing plant.

3.1.1 Commercial nest-run shell eggs. When commercial nest-run shell eggs are to be used in the processing of kind (a) product, the following conditions shall be met:

a. A certificate of conformance by the processor, including approved USDA labeling, shall indicate that the product was produced from high quality shell eggs suitable for table grade use. These eggs shall consist of straight run lots, i.e. no prior segregation other than removal of some obvious undergrades at the time of gathering.

b. Shell eggs shall be collected from farms and delivered directly to the breaking plant or where lots are assembled by and received from dealers. Both sources shall have refrigeration facilities for holding and shipping the eggs at 60°F (16°C) or below.

c. A complete set of records and a producer lot identification system shall be maintained regarding the quantity of eggs accepted from each source for breaking, including also a quality record for each lot of shell eggs (i.e., rejection on the basis of air temperature of delivery vehicles, condition as to age since eggs were laid, inedibles, loss, leakers, dirties, blood and meat spots, etc.).

d. All shell eggs to be used for breaking shall be washed and sanitized in accordance with USDA methods.

3.1.2 Grade A shell eggs. When graded shell eggs are used, they shall be covered by a grading certificate showing that the lot meets the requirements for US Grade A interior quality or better eggs. The grading certificate shall show that the eggs were graded within 10 days prior to the date of breaking.

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3.2 Processing. Processing shall be in accordance with table II.

TABLE II. Processing requirements

Item	Requirement
1	The eggs shall be handled and processed in compliance with the requirements described in the USDA Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59).
2	Sugar and salt, as applicable, shall be added prior to pasteurization.

3.3 End item requirements. The egg and egg products shall meet the physical, analytical and microbiological requirements of tables III and IV.

TABLE III. Physical requirements

Item	Requirement
1	The products shall be free from blood spots, meat spots, shell particles, paper and packaging materials, and other extraneous foreign material.
2	There shall be no off odors such as sour, musty, metallic, putrid, or other objectionable odors.
3	There shall be no visible egg yolk in kind (c).
4	Temperature at destinations shall be not more than 10°F (-12°C) and shall show no evidence of thawing and refreezing.

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TABLE IV. Analytical and microbiological requirements (see 6.1)

Item	Test	Frozen table grade whole eggs	Frozen whole eggs	Frozen white	Frozen yolk	Frozen sugar yolk	Frozen salt yolk
1	Egg solids, not less than, percent	24.5	24.5	11.5	43.0	38.7 1/	38.7 1/
2	Total solids, not less than, percent	24.5	24.5	11.5	43.0	48.5	48.5
3	Sugar, percent	---	---	---	---	10.0 2/	---
4	Salt, percent	---	---	---	---	---	10.0 2/
5	Aerobic plate count, not more than, per gram 3/	15,000	50,000	50,000	50,000	50,000	50,000
6	Yeast and mold count, not more than, per gram 3/	50	50	50	50	50	50
7	Salmonellae test 3/	4/	4/	4/	4/	4/	4/

1/ Forty-three percent on a sugar-free or salt-free basis, as applicable.

2/ Tolerance: Plus or minus 0.2 percent.

3/ Samples for microbiological examination shall arrive at the Government laboratory in a solidly frozen condition and held in such condition until tested.

4/ Shall be negative.

3.4 Plant qualification. The product shall originate, be processed, marked, and handled in plants which are operated under the Regulations Governing the Inspection of Eggs and Egg Products.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection (plant sanitation inspection).

4.2.1 Civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.2 Military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.4.

4.3 Acceptance inspection.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the US Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for military agencies (see 4.3.4)).

4.3.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.3 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.3.1 Examination of sugar and salt. Examination shall be made of labels, invoices, or other valid documents to determine conformance to the identity requirements specified in items 3 and 4, table I. Nonconformance to one or more identity requirements shall be cause for rejection of the involved component lot or finished product made therefrom, as applicable.

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4.3.3.2 Examination of eggs. Examination of eggs for compliance with requirements of table I (1) shall be ascertained by labels or other valid documents. For table grade, type II, kind (a), each lot offered by the supplier for breaking shall be examined to determine compliance with US Grade A interior quality requirements. Minimum sample size for number of cases of shell eggs in each lot to be inspected for determination of quality shall be as specified in the USDA publication referenced in 3.1.

4.3.3.3 Testing of components. Testing of components shall be as shown in table V. Results shall apply to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE V. Testing of components (see 4.3.3)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to nearest	Test reference and requirement paragraph
Raw liquid eggs or egg products for type II kind (a) only	One pint (473 ml)	Pounds (kg)	1/	Aerobic plate count	2/	4.5.2 3.1
Five, 10-lb. and 30-lb. (2.3, 4.5 and 13.6 kg) cans 3/	One can w/lid	Cans	S-1	Tin plate	0.01 lb. /base box	4.5.3 5.1.1.1 5.1.1.2
30-lb. (13.6 kg) cans 3/	One end cover and one 4 inches by 4 inches (102 by 102 mm) panel from side	Cans	S-1	Base weight plate	lb/base box	4.5.7 5.1.1.2 5.1.1.2.1

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TABLE V. Testing of components (see 4.3.3) (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to nearest	Test reference and requirement paragraph
Four, 5, 8 and 10-lb. (1.8, 2.3, 3.6 and 4.5 kg) cartons	One carton	Cartons	S-1	Basis weight carton stock	lb/1000 sq. ft. (g per sq. meters)	4.5.4 5.1.1.3 Table XI
				Thickness carton stock after coating	0.001 in. (0.01 mm)	4.5.5 5.1.1.3 Table XI
				Body seam heat seal	Pass or fail	4.5.6 5.1.1.3

1/ One sample of raw liquid eggs or egg product shall be drawn aseptically from each batch before processing until all batches making up a single lot have been sampled and composited. Cumulative sample shall be kept under 40°F (4°C) and after completion of the composite, shall be frozen promptly and handled in accordance with 4.4.

2/ In accordance with referenced test procedure.

3/ Exterior can coating, when required, and can label, shall be tested in accordance with the applicable subsidiary specifications except that tests shall be performed on the same cans submitted for tinplate testing.

4.3.3.4 In-process examination. Examination shall be performed during handling and processing to determine conformance to the referenced requirements in 3.2, as concerns methods, times and temperature controls during processing and storage. Time and temperature records shall be maintained. Nonconformance to one or more requirements reflected by actual examination or by records shall be cause for rejection of involved quantity of the finished product.

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4.3.4 Examination of end item. Classification of defects found during examination shall be in accordance with tables VII and VIII and 4.3.4.1 through 4.3.4.7. The examination criteria for the above referenced tables shall be as shown in table VI. The acceptable quality levels (AQLs) for table VII shall be expressed in terms of percent defective and in terms of defects per hundred units for table VIII.

TABLE VI. End item examination criteria 1/

Table	Inspection level	Sample unit	Lot size expressed in	AQLs	
				Major	Minor
VII	S-3	One primary container	Primary containers	---	2.5
VIII	S-3	One primary container	Primary containers	1.0	---

1/ Sampling shall be accomplished any time after the product has been frozen solidly to a temperature of 10°F (-12°C) within a period of 60 hours.

TABLE VII. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight less than specified minus tolerance (see tables IX and X). 2/

1/ The lot shall be rejected if the average net weight of the samples is less than the net weight specified.

2/ Report to the nearest weight.

Cans/cartons (as applicable)	Nearest weight
5 lbs (2.3 kg) or less	1/16-ounce (2 g)
Over 5 to 10 lbs (2.3 to 4.5 kg)	1/4-ounce (7 g)
Over 10 pounds (4.5 kg)	1/2-ounce (14 g)

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TABLE VIII. Examination of product characterisitics 1/

Category	Defect
Major	
151	Presence of extraneous material, e.g., shell particles, blood spots, meat spots, or other defects normally associated with the product.
152	Evidence of thawing and refreezing.
153	Presence of visible amounts of egg yolk in kind (c).

1/ Presence of foreign material, e.g., metal, glass, wood, insect, insect fragments, paper and packaging materials, etc., or foreign flavors or odors, e.g., stale, strong eggy, putrid, sour, musty, metallic, etc., shall be cause for rejection of the lot. Observations shall be made when product is thawed.

4.3.4.1 Examination of labeling of primary containers. Examination of can labeling for military agencies shall be in accordance with the examination criteria of MIL-L-1497 and the same sampling plan shall be acceptable when examining the labeling of paperboard milk cartons (when used). A defect shall be scored if labeling is missing, incorrect, or illegible. For civil agencies, labeling, not in accordance with 5.3.1.1, shall not be acceptable.

4.3.4.2 Examination of can coating. Examination of can coating, when applicable, shall be in accordance with the examination criteria of TT-C-495.

4.3.4.3 Examination of cans. The examination of level B cans for condition and seam leakage shall be in accordance with applicable examination criteria in PPP-C-29, except that the defects for springer, flipper or swell and key shall not apply.

4.3.4.4 Examination of polyethylene coating. Conformance of polyethylene coating thickness to requirements of table XI shall be ascertained by examination of invoices or other valid documentation from the supplier of the carton material. Nonconformance shall indicate unacceptable material and use of such shall be cause for rejection of the involved end item.

4.3.4.5 Examination of shipping containers. Examination of the filled and closed shipping container for level B shall be in accordance with the examination criteria contained in the appendix to PPP-B-636, or PPP-B-1163 as applicable. In addition, the following defects shall be included in the table of examination: Major - Marking missing, incorrect or illegible. Minor (1) Arrangement not

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as specified, net weight not as specified; (2) As applicable, missing full height fiberboard partitions. Also, when applicable, for level B pack, shipping containers reinforced with other than nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape, shall be unacceptable. For level C pack examination shall be in accordance with aforementioned specification, except that only the additional defects and closure requirements (see 5.2.2) shall apply.

4.3.4.6 Examination of storage time. Determination for classification of product as type I or II shall be made by an examination of labels, storage charts, markings, or other valid documents. Product exceeding storage time limitations shall be cause for rejection of the lot.

4.3.4.7 Examination of destination temperatures. The sample unit shall be one primary container for the 5 and 10-pound (2.3 and 4.5 kg) cans and the paperboard cartons, and one shipping container for the 30-pound (13.6 kg) can. Lot size shall be expressed in terms of the sample unit, and sample size for inspection shall be the number of sample units indicated by inspection level S-2. At time of product receipt, the maximum internal temperature of the product taken shall be in accordance with the requirement in table III, item 4. Nonconformance to the referenced temperature requirement or any evidence of thawing or refreezing shall be cause for rejection of the lot.

4.4 Sampling procedure and acceptance criteria for testing of finished products (see 3/ table IV). The finished product shall be tested for egg solids, total solids, sugar, salt, aerobic plate count, yeast and mold count and Salmonella as specified in table IV. Procedures for testing shall be in accordance with 4.5. Test requirements for Salmonella, aerobic plate count and yeast and mold count shall be on a unit basis. Test requirements for all other requirements shall be on a lot average basis. The sample unit for Salmonella, total plate count and yeast and mold count shall be an 8-ounce (227 g) sample derived aseptically from one primary container. Lot size shall be expressed in terms of the sample unit. The sample size shall be the number of cans indicated by inspection level S-1. The sample for egg solids, total solids, sugar, and salt shall be a 1-pound (0.45 kg) composite derived from the number of containers used for the microbiological requirements. The sample for testing may be drawn in the liquid state prior to freezing. Salmonella shall be reported on a pass or fail basis. Microbiological counts shall be reported in accordance with Compendium of Methods for the Microbiological Examination of Foods. Egg solids, total solids, sugar and salt shall be reported to the nearest 0.1 percent. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

4.5 Test methods. Test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

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4.5.1 Analysis shall be made in accordance with Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Egg and Egg Products:

Item	Test 1/	Section	Method
1	Egg solids 2/	Total solids	Vacuum method
2	Total solids	Total solids	Vacuum method
3	Salt (chloride)	Chloride	Method II
4	Sugar	Glucose and sucrose	Dried eggs

1/ Prepare sample in accordance with the method outlined in Compendium of Methods for the Microbiological Examination of Foods of the American Public Health Association, Chapter: Eggs and Egg Products, Section: Frozen Eggs.

2/ Frozen salt yolk and frozen sugar yolk - subtract percent salt or sugar from percent total solids.

4.5.2 Microbiological analyses. The aerobic plate count and the yeast and mold count shall be determined in accordance with the Compendium of Methods for the Microbiological Examination of Foods, Chapter: Egg and Egg Product, Section: Frozen Eggs:

Item	Test	Method
1	Aerobic count	Aerobic count
2	Yeast and mold count	Yeast and mold

4.5.2.1 Salmonella shall be determined in accordance with the USDA Laboratory Methods for Egg Products.

4.5.3 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

4.5.4 Basis weight test. Basis weight shall be tested in accordance with method 5022 of FED-STD-101.

4.5.5 Thickness of carton stock. Thickness of carton stock after coating shall be measured in accordance with method 1003 of FED-STD-101.

4.5.6 Carton body seam heat seal. The carton body seam shall be examined for heat seal separation at point of contact after placing the carton in a cold box with a temperature of minus 20°F ± 2°F (-29 ± 1°C) for a minimum of two hours.

4.5.7 Base plate weight. Base plate weight shall be determined in accordance with QQ-T-425.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level B or level C, as specified (see 6.1).

5.1.1 Level B. The product shall be packaged in accordance with 5.1.1.1, 5.1.1.2 or 5.1.1.3, as specified (see 6.1).

5.1.1.1 Five and 10-pound (2.3 and 4.5 kg) cans. The product shall be packaged in a new, round, full opening, slip-cover style, metal can (see table IX). As specified (see 6.1), the size of the can, the net weight and allowable unit variation shall be as specified in table IX. The cans shall be made throughout from not less than commercial 0.25 pound electrolytic tin plate per base box (see 4.5.3). The inside of the can and cover shall be coated with an enamel supplied by the manufacturer for the specific product being packaged. The can and cover shall be coated overall outside with a coating conforming to type I or when specified (see 6.1) type III of TT-C-495. When specified (see 6.1), cans with or without commercial coating will be acceptable.

5.1.1.2 Thirty-pound (13.6 kg) cans. Thirty-pounds (13.6 kg) of the product shall be packaged in a new, round, full opening, slip-cover style, tapered or straight side metal can (see table IX). The body and end of the can shall be made from not less than 100-pound nominal base weight plate and shall be corrug or beaded for additional rigidity and the slip covers may be made from not less than 80-pound base weight plate. The side seam shall be soldered or welded, the bottom shall be cement lined and double seamed, and the can rim shall be curled outward. When side seams are welded, the entire interior seam of the container shall be covered with enamel. The slip cover shall be snug-fitting and shall have a flange not less than 11/16 inch (18 mm) long with a curled edge. The can shall be made throughout from not less than commercial 0.25 pound electrolytic tin plate per base box. The inside of the can and cover shall be coated with an enamel supplied by the manufacturer for the specific product being packaged. The can and cover shall be coated overall outside as specified in 5.1.1.1.

5.1.1.2.1 Alternatively, (see 6.1) the can shall be made as specified in 5.1.1.2, except that body shall be made from not less than 107-pound nominal base weight plate, the ends may be made from 107-pound or 100-pound nominal base weight plate, but the 100-pound bottoms shall be corrugated or beaded for additional rigidity, and the bottom seams shall be compound-lined.

5.1.1.3 Cartons. The product shall be packaged in polyethylene coated paperboard milk cartons. As specified (see 6.1), the net weight, carton volume and allowable unit variation shall be as specified in table X. The cartons shall be fabricated from folding grade bleached sulfate paperboard which conforms to the physical requirements of table XI. Cartons shall be tuck bottom (corner side seam) pitcher-pour style with the machine direction of the paperboard running perpendicular to the score line of the carton opening The body seam shall be securely bonded by a coating-to-coating heat seal the full depth of the carton.

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5.1.2 Level C. The product shall be packaged to afford adequate protection against deterioration and damage during shipment from supply source to the first consuming activity. The supplier may use his standard practice when it meets these requirements.

TABLE IX. Can sizes, net weight per can, allowable unit variation and number and arrangement of cans in each shipping container (see 6.1)

Can size	Net weight (pounds or kg)	Allowable unit variations	Number in cases	Length	Width	Depth
601 by 604	5 (2.3 kg)	-2 percent	6	3	2	1
610 by 500		-2 percent				
702 by 804	10 (4.5 kg)	-2 percent	4	2	2	1
914 by 1210	30 (13.6 kg)	-1 percent	1	1	1	1
or 1300	30 (13.6 kg)	-1 percent	1	1	1	1
1000 by 1209	30 (13.6 kg)	-1 percent	1	1	1	1
Tapered can Top O.D. - 1007 Bottom O.D. - 911 Height (overall cover on) 1214 or 1300						

TABLE X. Net weight per carton, allowable unit variation, carton volume, number and arrangement of cartons in shipping container

Net weight per carton (pounds) (kg)	Allowable unit variations	Carton volume	Number cartons per case	Packing arrangement		
				Length	Width	Depth
4 (1.8 kg)	-2 percent	US Half Gallon (1.9 liter)	6	3	2	1
8 (3.6 kg)	-2 percent	US Gallon (3.8 liter)	4	2	2	1
5 (2.3 kg)	-2 percent	Imperial Half Gallon (2.4 liter)	6	3	2	1
10 (4.5 kg)	-2 percent	Imperial Gallon (4.7 liter)	4	2	2	1

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TABLE XI. Physical requirements for polyethylene coated paperboard cartons

Net weight per carton (pounds) (kg)	Basis Weight (Pounds per 1000 square ft.) (g per sq meter)	Thickness after coating (inches) (mm)	Polyethylene coating thickness (inches) (mm) Outside	Inside
4 (1.8)	94.3 (460)	0.022 (0.6)	0.0005 (0.01)	0.00075 (0.019)
8 (3.6)	110 (537)	0.026 (0.7)	0.0005 (0.01)	0.00075 (0.019)
5 (2.3)	94.3 (460)	0.022 (0.6)	0.0005 (0.01)	0.00075 (0.019)
10 (4.5)	110 (537)	0.026 (0.7)	0.0005 (0.01)	0.00075 (0.019)

5.2 Packing. The product, packaged as specified in 5.1, shall be packed in accordance with level B or C, as specified (see 6.1). The cans of product, in the number and arrangement specified in table IX, shall be packed on end in snug-fitting shipping containers in accordance with level B. Not more than 40 pounds (18.1 kg) of the product, plus one percent tolerance, packaged as specified in 5.1.2, shall be packed in accordance with level C (see 6.1). The cartor of product shall be packed in a snug-fitting shipping container in accordance with level B or C as specified (see 6.1). For level B, the cartons of product shall be packed on their side in the number and arrangement specified in table X For level C packing, the cartons may be packed either on end or on their side, in the number and arrangement commercially used by the supplier. When the cartons are packed on end, the shipping container shall be provided with full height fiberboard partitions. Not more than 40 pounds (18.1 kg) of the product packaged as specified in 5.1.1.3, shall be packed in accordance with level C.

5.2.1 Level B. The shipping container shall be a fiberboard box, constructed closed and reinforced in accordance with style RSC, V3c or V3s, or V4s of PPP-B-636. Alternatively, the shipping container shall be a wax-impregnated box constructed in accordance with style RSC, class I, type 275 of PPP-B-1163. The wax-impregnated container shall be closed with hot melt glue and reinforced with two lengthwise and one girthwise band.

5.2.1.1 When specified (see 6.1 and 6.2), the shipping container specified in 5.2.1 shall be reinforced with non-metallic strapping or pressure sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636

5.2.2 Level C. The shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable, except the closure of the container shall be in accordance with the appendix of PPP-B-636.

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5.3 Labeling and marking.

5.3.1 Civil agencies.

5.3.1.1 Unit containers. Commercial labeling meeting the requirements of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder will be acceptable.

5.3.1.2 Shipping containers. Shipping containers shall be marked in accordance with FED-STD-123.

5.3.1.2.1 Additional marking. In addition to the marking specified in 5.3.1.2, the following information shall be stenciled or printed on top of the shipping container in letters not less than 1-1/2 inches (38 mm) high:

KEEP FROZEN
(0°F (-18°C) or below)
THIS END UP

5.3.2 Military agencies.

5.3.2.1 Cans. Cans shall be labeled in accordance with MIL-L-1497 and shall include the following:

Name and type of product (in letters larger than any others used for the label).

Net weight.

Name and address of manufacturer (plant location).

Date packed (day, month, year).

The following statement, except for type II, kind (a), shall be included (in letters as large as used for the name of the product):

USE FOR BAKED GOODS ONLY

5.3.2.2 Cartons. Cartons shall be labeled in accordance with 5.3.1.1.

5.3.2.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. Alternatively, when specified, commercial markings plus any specified markings required by the contract or order, will be acceptable. In addition, the following information shall be stenciled or printed on top of the shipping container in letters not less than 1-1/2 inches (38 mm) high:

KEEP FROZEN
(0°F (-18°C) or below)
THIS END UP

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and kind required (see 1.2).
(c) When analytical requirements are not needed (see table IV).
(d) Type of can coating (see 5.1.1.1).
(e) When cans without commercial can coating will be acceptable (see 5.1.1.1).
(f) Whether cans or cartons are required for level B packaging and the size of level B packaging required (see 5.1.1 and 5.1.1.3).
(g) Determination of can size for 5, 10, or 30-pound (2.3, 4.5 or 13.6 kg) container (see table IX).
(h) Levels of packaging and packing required (see 5.1 and 5.2).
(i) When packing specified in 5.2.1.1.

6.2 Intended use. The products covered by this specification are intended for domestic and overseas use. Table grade frozen egg type II, kind (a), only, is primarily intended for preparing scrambled eggs, omelets, western sandwiches etc., where only high quality product should be used. All other items, including types I and II, kinds (b) through (f), are intended for use in baking and cooking by Federal agencies and by suppliers in preparing subsistence items for use by the Armed Forces.

6.2.1 Based on conditions known or expected to be encountered during shipping, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.14D/DSAR 4145.7.

6.3 Packing specified in 5.2.1.1 is intended for transfer at sea operations or specific overseas operations.

6.4 Destination inspection (for civil agencies). When the finished product has been inspected and passed at a point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

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C-E-230E

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
HEW-FDA, NIH
USDA-AMS
VA-DMS

Project No. 8910-0369

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other referenced documents herein. Price 1.00 cents each.

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